POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5011 | Semester V | Theory | 4 Credits

Construction Management and Safety Engineering

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Civil Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5011 as Construction Management and Safety Engineering in Semester V for Civil Engineering. The official SITTTR detailed source was unavailable during this cycle. With user approval, explanatory coverage is based on reliable public diploma construction-management curriculum material and is not represented as recovered official SITTTR Course 5011 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders. This handbook supports study only; all construction decisions and site activities require approved drawings, applicable law, competent supervision, permits and site-specific safety planning.

Verified source note: The official detailed source was inaccessible. The public diploma source used for construction planning, site organisation, quality control and safety is linked in References.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Construction Management and Safety Engineering develops organised, safety-first practice for planning, coordinating and controlling construction work. It connects scope, resources, schedules, quality, cost, site organisation, risk controls and professional responsibility.

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Prerequisite foundation

Building-construction fundamentals, communication, basic arithmetic and a strict commitment to safety.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain construction-project roles, resources, planning stages and site organisation.
2. Use basic scheduling, progress-control and resource-planning concepts.
3. Explain quality control, cost awareness and responsible documentation.
4. Apply construction-safety risk controls and incident-prevention principles to a site scenario.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain construction project lifecycle, stakeholders, resources and site organisation.	Understand
CO2	Prepare and interpret basic work-breakdown, bar-chart or network-based planning information.	Apply
CO3	Explain quality, progress, cost and documentation controls for construction work.	Understand
CO4	Apply the hierarchy of safety controls to typical construction hazards.	Apply

Official CO-PO articulation matrix							
CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7-PO11**
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Construction Planning and Organisation	Project lifecycle, stakeholders, resources, planning, organisation, site layout and communication.	Approx. 15 hours	CO1	Module 1
2	Scheduling, Resources and Progress Control	WBS, activity sequencing, bar charts, CPM/PERT concepts, critical path, resource planning and progress measurement.	Approx. 16 hours	CO2	Module 2
3	Quality, Cost and Contract Awareness	Inspection/testing, quality plans, material control, cost/time awareness, measurement, change documentation and ethical professional practice.	Approx. 15 hours	CO3	Module 3
4	Construction Safety Engineering	Hazard identification, risk assessment, hierarchy of controls, PPE, permits, excavation, height, lifting, electrical and emergency practices.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Construction Planning and Organisation

Project lifecycle, stakeholders, resources, planning, organisation, site layout and communication.

CO1 · Approx. 15 hours

Scheduling, Resources and Progress Control

WBS, activity sequencing, bar charts, CPM/PERT concepts, critical path, resource planning and progress measurement.

CO2 · Approx. 16 hours

Quality, Cost and Contract Awareness

Inspection/testing, quality plans, material control, cost/time awareness, measurement, change documentation and ethical professional practice.

CO3 · Approx. 15 hours

Construction Safety Engineering

Hazard identification, risk assessment, hierarchy of controls, PPE, permits, excavation, height, lifting, electrical and emergency practices.

CO4 · Approx. 14 hours

MODULE 1

Construction Planning and Organisation

Project lifecycle, stakeholders, resources, planning, organisation, site layout and communication.

Approx. 15 hours **CO1** **Understand · Apply**

Learning goals

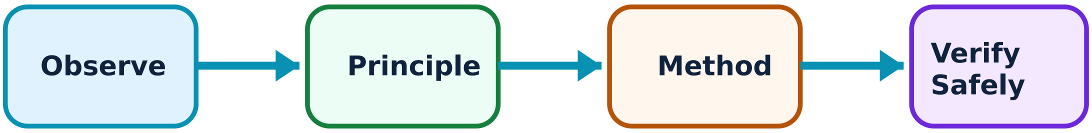
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Construction Planning and Organisation: identify the system, state its principle, follow the correct method and verify the result safely.

1. Project lifecycle and stakeholders−

Construction moves from concept, design and procurement through execution, control, handover and closeout. Owners, designers, contractors, supervisors, workers, suppliers and authorities have distinct duties and interfaces.

Study and application method

Map a project stage to key deliverables, responsible role, information input and approval/inspection point.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Map a project stage to key deliverables, responsible role, information input and approval/inspection point.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not start work from incomplete verbal instructions; confirm current approved drawings, scope and authority.

2. Resources and site organisation−

Labour, materials, plant, money, time and information must be coordinated. Site layout affects material flow, access, storage, lifting, welfare, emergency response and security.

Study and application method

Sketch a simple site plan showing access, storage, plant zones, utilities, waste, pedestrian routes and emergency arrangements.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch a simple site plan showing access, storage, plant zones, utilities, waste, pedestrian routes and emergency arrangements.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Never place storage or plant where it blocks safe access, egress, services or lifting operations.

3. Communication and records−

Daily records, drawings, specifications, requests, inspections and progress reports create traceability and support coordinated decisions.

Study and application method

Use dated, factual records with location, activity, quantities, condition, instruction source and photographs only where permitted.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use dated, factual records with location, activity, quantities, condition, instruction source and photographs only where permitted.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not alter records, conceal deviations or rely on outdated document revisions.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Scheduling, Resources and Progress Control

WBS, activity sequencing, bar charts, CPM/PERT concepts, critical path, resource planning and progress measurement.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

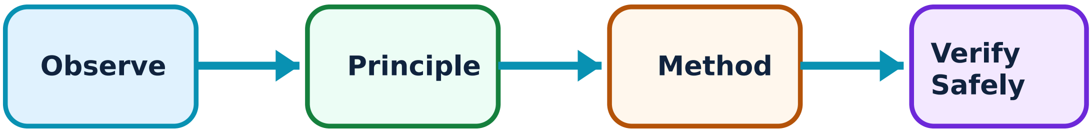
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Scheduling, Resources and Progress Control: identify the system, state its principle, follow the correct method and verify the result safely.

1. Work breakdown and activity logic−

A work breakdown structure decomposes scope into manageable deliverables and activities. Logical dependencies show what must finish/start before another activity can proceed.

Study and application method

Define deliverable, activities, predecessor/successor, duration basis and responsible resource; review for missing interfaces.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define deliverable, activities, predecessor/successor, duration basis and responsible resource; review for missing interfaces.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not treat a schedule as a list of dates—unresolved dependencies and constraints make dates unreliable.

2. Bar charts and network methods−

Bar charts communicate planned activity timing; network methods reveal dependencies, float and critical path. They assist planning but must be updated with real progress and constraints.

Study and application method

Create a small precedence network, calculate/identify critical sequence conceptually and compare planned versus actual progress.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Create a small precedence network, calculate/identify critical sequence conceptually and compare planned versus actual progress.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A critical-path result is only as valid as activity logic, duration estimates and update data.

3. Resource and progress control−

Progress control compares actual quantities/time/cost with plan, identifies variance and supports corrective action. Resource loading must consider availability, productivity, safety and space.

Study and application method

Set baseline, collect measurable progress, calculate variance, identify cause and propose a safe feasible corrective action with owner/date.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Set baseline, collect measurable progress, calculate variance, identify cause and propose a safe feasible corrective action with owner/date.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not recover schedule by unsafe acceleration, excessive overtime or skipping quality checks.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Quality, Cost and Contract Awareness

Inspection/testing, quality plans, material control, cost/time awareness, measurement, change documentation and ethical professional practice.

Approx. 15 hours | **CO3** | **Understand · Apply**

Learning goals

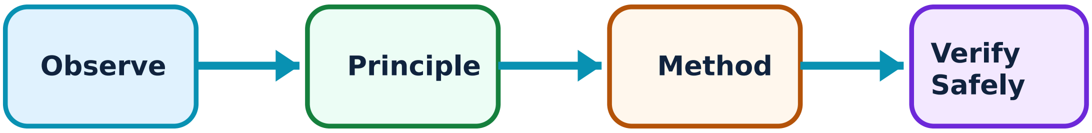
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Quality, Cost and Contract Awareness: identify the system, state its principle, follow the correct method and verify the result safely.

1. Quality control and inspection−

Quality control verifies that work/materials meet specified requirements through planned inspections, tests, hold points and records. Early prevention is cheaper than late correction.

Study and application method

For an activity, identify acceptance criteria, inspection stage, required record, responsible person and treatment of nonconformance.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For an activity, identify acceptance criteria, inspection stage, required record, responsible person and treatment of nonconformance.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not cover, conceal or continue work beyond a hold point without required inspection approval.

2. Cost and productivity awareness−

Cost includes materials, labour, plant, overhead, rework and time-related impacts. Productivity must be interpreted with quality, safety and realistic resource conditions.

Study and application method

Record quantities, resources and constraints transparently; compare actual to baseline and distinguish direct cause from assumption.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Record quantities, resources and constraints transparently; compare actual to baseline and distinguish direct cause from assumption.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not reduce cost by substituting unapproved material or eliminating required safety/quality controls.

3. Changes and professional ethics−

Variations, claims and changed conditions require timely notification, documentation and authorised decision. Integrity protects workers, clients and the public.

Study and application method

Document condition, reference drawing/specification, proposed impact and approval path before implementing a change.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Document condition, reference drawing/specification, proposed impact and approval path before implementing a change.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Never approve work outside delegated authority or manipulate measurements/certificates.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Construction Safety Engineering

Hazard identification, risk assessment, hierarchy of controls, PPE, permits, excavation, height, lifting, electrical and emergency practices.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

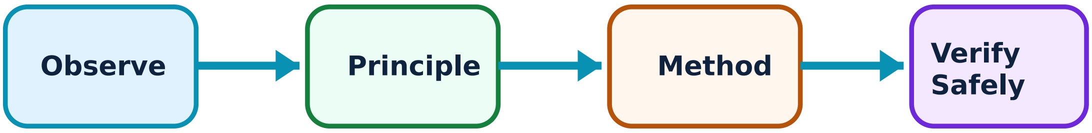
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Construction Safety Engineering: identify the system, state its principle, follow the correct method and verify the result safely.

1. Hazard and risk assessment−

A hazard can cause harm; risk combines likelihood and consequence. Risk assessment identifies work steps, hazards, exposed people, controls and residual risk.

Study and application method

Break work into steps, identify hazards, apply elimination/substitution/engineering/administrative/PPE controls and assign verification responsibility.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Break work into steps, identify hazards, apply elimination/substitution/engineering/administrative/PPE controls and assign verification responsibility.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Risk forms are not paperwork alone; stop and reassess whenever conditions change.

2. High-risk construction activities−

Excavation, work at height, scaffolding, lifting, temporary works, electrical work and demolition require specific planning, inspection, competence and permits.

Study and application method

For a task, identify exclusion zones, load/ground/support checks, competent-person inspection, communication and emergency plan.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For a task, identify exclusion zones, load/ground/support checks, competent-person inspection, communication and emergency plan.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Never enter an unsupported excavation, work on an unprotected edge or stand under a suspended load.

3. Incident prevention and emergency response−

Safety culture encourages reporting of hazards/near misses, learning and corrective action. Emergency readiness includes alarm, communication, first aid, evacuation and rescue arrangements.

Study and application method

State alarm route, muster point, responsible contacts, required equipment and actions for a realistic scenario.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Do not attempt rescue beyond training/equipment; summon competent emergency support promptly.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Construction Planning and Organisation

Use the concept flow to revise observation, principle, method and verification.

Module 2: Scheduling, Resources and Progress Control

Use the concept flow to revise observation, principle, method and verification.

Module 3: Quality, Cost and Contract Awareness

Use the concept flow to revise observation, principle, method and verification.

Module 4: Construction Safety Engineering

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Prepare a WBS and bar chart for a small instructor-provided construction activity.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Create a basic site-layout sketch with material flow and emergency access.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Develop an inspection checklist for a simulated construction operation.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Complete a risk assessment for a work-at-height or excavation scenario using the hierarchy of controls.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	**Official mark / conversion**	**Student evidence**
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Construction Planning and Organisation

Explain Project lifecycle and stakeholders and state one correct method, application or precaution.−

Construction moves from concept, design and procurement through execution, control, handover and closeout. Owners, designers, contractors, supervisors, workers, suppliers and authorities have distinct duties and interfaces.

Answer framework: Map a project stage to key deliverables, responsible role, information input and approval/inspection point.

Important point: Do not start work from incomplete verbal instructions; confirm current approved drawings, scope and authority.

Explain Resources and site organisation and state one correct method, application or precaution.−

Labour, materials, plant, money, time and information must be coordinated. Site layout affects material flow, access, storage, lifting, welfare, emergency response and security.

Answer framework: Sketch a simple site plan showing access, storage, plant zones, utilities, waste, pedestrian routes and emergency arrangements.

Important point: Never place storage or plant where it blocks safe access, egress, services or lifting operations.

Explain Communication and records and state one correct method, application or precaution.−

Daily records, drawings, specifications, requests, inspections and progress reports create traceability and support coordinated decisions.

Answer framework: Use dated, factual records with location, activity, quantities, condition, instruction source and photographs only where permitted.

Important point: Do not alter records, conceal deviations or rely on outdated document revisions.

Module 2 — Scheduling, Resources and Progress Control

Explain Work breakdown and activity logic and state one correct method, application or precaution.−

A work breakdown structure decomposes scope into manageable deliverables and activities. Logical dependencies show what must finish/start before another activity can proceed.

Answer framework: Define deliverable, activities, predecessor/successor, duration basis and responsible resource; review for missing interfaces.

Important point: Do not treat a schedule as a list of dates—unresolved dependencies and constraints make dates unreliable.

Explain Bar charts and network methods and state one correct method, application or precaution.−

Bar charts communicate planned activity timing; network methods reveal dependencies, float and critical path. They assist planning but must be updated with real progress and constraints.

Answer framework: Create a small precedence network, calculate/identify critical sequence conceptually and compare planned versus actual progress.

Important point: A critical-path result is only as valid as activity logic, duration estimates and update data.

Explain Resource and progress control and state one correct method, application or precaution.−

Progress control compares actual quantities/time/cost with plan, identifies variance and supports corrective action. Resource loading must consider availability, productivity, safety and space.

Answer framework: Set baseline, collect measurable progress, calculate variance, identify cause and propose a safe feasible corrective action with owner/date.

Important point: Do not recover schedule by unsafe acceleration, excessive overtime or skipping quality checks.

Module 3 — Quality, Cost and Contract Awareness

Explain Quality control and inspection and state one correct method, application or precaution.−

Quality control verifies that work/materials meet specified requirements through planned inspections, tests, hold points and records. Early prevention is cheaper than late correction.

Answer framework: For an activity, identify acceptance criteria, inspection stage, required record, responsible person and treatment of nonconformance.

Important point: Do not cover, conceal or continue work beyond a hold point without required inspection approval.

Explain Cost and productivity awareness and state one correct method, application or precaution.−

Cost includes materials, labour, plant, overhead, rework and time-related impacts. Productivity must be interpreted with quality, safety and realistic resource conditions.

Answer framework: Record quantities, resources and constraints transparently; compare actual to baseline and distinguish direct cause from assumption.

Important point: Do not reduce cost by substituting unapproved material or eliminating required safety/quality controls.

Explain Changes and professional ethics and state one correct method, application or precaution.−

Variations, claims and changed conditions require timely notification, documentation and authorised decision. Integrity protects workers, clients and the public.

Answer framework: Document condition, reference drawing/specification, proposed impact and approval path before implementing a change.

Important point: Never approve work outside delegated authority or manipulate measurements/certificates.

Module 4 — Construction Safety Engineering

Explain Hazard and risk assessment and state one correct method, application or precaution.−

A hazard can cause harm; risk combines likelihood and consequence. Risk assessment identifies work steps, hazards, exposed people, controls and residual risk.

Answer framework: Break work into steps, identify hazards, apply elimination/substitution/engineering/administrative/PPE controls and assign verification responsibility.

Important point: Risk forms are not paperwork alone; stop and reassess whenever conditions change.

Explain High-risk construction activities and state one correct method, application or precaution.−

Excavation, work at height, scaffolding, lifting, temporary works, electrical work and demolition require specific planning, inspection, competence and permits.

Answer framework: For a task, identify exclusion zones, load/ground/support checks, competent-person inspection, communication and emergency plan.

Important point: Never enter an unsupported excavation, work on an unprotected edge or stand under a suspended load.

Explain Incident prevention and emergency response and state one correct method,

application or precaution.−

Safety culture encourages reporting of hazards/near misses, learning and corrective action. Emergency readiness includes alarm, communication, first aid, evacuation and rescue arrangements.

Answer framework: State alarm route, muster point, responsible contacts, required equipment and actions for a realistic scenario.

Important point: Do not attempt rescue beyond training/equipment; summon competent emergency support promptly.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Construction Planning and Organisation.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Scheduling, Resources and Progress Control.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Quality, Cost and Contract Awareness.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Construction Safety Engineering.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Project lifecycle, stakeholders, resources, planning, organisation, site layout and communication..
2. Develop a complete answer or practical plan for: WBS, activity sequencing, bar charts, CPM/PERT concepts, critical path, resource planning and progress measurement..

3. Develop a complete answer or practical plan for: Inspection/testing, quality plans, material control, cost/time awareness, measurement, change documentation and ethical professional practice..
4. Develop a complete answer or practical plan for: Hazard identification, risk assessment, hierarchy of controls, PPE, permits, excavation, height, lifting, electrical and emergency practices..

LAST-MILE REVISION

Quick Revision

Module 1: Construction Planning and Organisation

Project lifecycle, stakeholders, resources, planning, organisation, site layout and communication.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Scheduling, Resources and Progress Control

WBS, activity sequencing, bar charts, CPM/PERT concepts, critical path, resource planning and progress measurement.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Quality, Cost and Contract Awareness

Inspection/testing, quality plans, material control, cost/time awareness, measurement, change documentation and ethical professional practice.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Construction Safety Engineering

Hazard identification, risk assessment, hierarchy of controls, PPE, permits, excavation, height, lifting, electrical and emergency practices.

- Match each topic to CO4.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Project lifecycle and stakeholders	Construction moves from concept, design and procurement through execution, control, handover and closeout.
Resources and site organisation	Labour, materials, plant, money, time and information must be coordinated.
Communication and records	Daily records, drawings, specifications, requests, inspections and progress reports create traceability and support coordinated decisions.
Work breakdown and activity logic	A work breakdown structure decomposes scope into manageable deliverables and activities.
Bar charts and network methods	Bar charts communicate planned activity timing; network methods reveal dependencies, float and critical path.
Resource and progress control	Progress control compares actual quantities/time/cost with plan, identifies variance and supports corrective action.
Quality control and inspection	Quality control verifies that work/materials meet specified requirements through planned inspections, tests, hold points and records.
Cost and productivity awareness	Cost includes materials, labour, plant, overhead, rework and time-related impacts.
Changes and professional ethics	Variations, claims and changed conditions require timely notification, documentation and authorised decision.
Hazard and risk assessment	A hazard can cause harm; risk combines likelihood and consequence.
High-risk construction activities	Excavation, work at height, scaffolding, lifting, temporary works, electrical work and demolition require specific planning, inspection, competence and permits.
Incident prevention and emergency response	Safety culture encourages reporting of hazards/near misses, learning and corrective action.

LEARNING RESOURCES

References

Recommended construction-management references

1. R. L. Peurifoy, Construction Planning, Equipment and Methods.
2. L. S. Srinath, PERT and CPM: Principles and Applications.
3. Applicable construction safety regulations and approved site procedures.

Approved public construction-management source

- https://hsbte.org.in/pdf/syllabus/CIVIL%20ENGG/6.pdf

This public source supports the study handbook while official Course 5011 content is unavailable; it does not replace approved project documents or safety procedures.